Adiabatically Bent Waveguides on Silicon Nitride Photonics for Compact and Dense Footprints

Jeong Hwan Song, Tangla D. Kongnyuy, Andim Stassen, Vignesh Mukund, and Xavier Rottenberg

***Abstract*— Low-loss bent waveguides drawn by an adiabatic profile with an offset at the interface between a bent and straight waveguides have been proposed, optimally designed, and experimentally evaluated for silicon nitride waveguide platforms, which compactness and high density are required. The profile is sinusoidal which can gradually increase the waveguide width up to 45° and then decrease the width up to 90° symmetrically. By employing the profile, modal confinement is increased and bending loss is decreased simultaneously. As a result of experimental evaluation, the characteristics of bending loss are improved. For instance, 0.215-dB/90° bending loss has been achieved at 5 μm radius. The loss of this bend is 0.4 dB/90° lower than that of a conventional bend. We also investigated bends when using a *C*-band light source theoretically. Less than 0.6 dB/90° bending loss has been shown using 3-D finite-difference time-domain simulations at 20 μm radius. Thus, it can be used in various applications using visible and infrared lights to reduce the device footprint.**

***Index Terms*— Adiabatic profile, bent waveguide, effective index, modal confinement, silicon nitride waveguide, silicon photonics.**

I. Introduction

THE demand for photonic integrated circuits (PICs) based on silicon nitride and silicon waveguides is increasing in various areas such as telecom [1], data-com [2], bio-medical [3], and image sensor [4] applications due to the compactness given by the high index contrast. Nevertheless, silicon waveguide remains transparent only for longer wavelength than 1.1μm [5], [6]. Therefore, the silicon based PICs are not available for the visible and near infrared light applications. On the other hand, PICs based on silicon nitride waveguide platforms show excellent optical properties in the wide wavelength range including visible and near infrared ranges [6], which are useful for biological and/or image sensor applications, and can also be fabricated by a complementary metal-oxide semiconductor (CMOS) compatible process.

One of key factors for high performance and compact PICs based on silicon nitride waveguide platforms is low bending losses with sharp bending radii. The index contrast of the silicon nitride waveguide is relatively smaller than that of a silicon on insulator (SOI) waveguide. Typical index contrast of the silicon nitride waveguide, defined as $\Delta = (n_{\text{core}}^2 - n_{\text{clad}}^2)/(2n_{\text{core}}^2)$, is ~20% at the wavelength of 638nm. For better understanding, a bending loss of less than 0.1dB/90° has been reported with a bending radius of 1μm on the conventional single mode waveguide of SOI using a 1550nm light source, where Δ is ~45% [7]. This index contrast difference makes the bending radius of silicon nitride waveguide to be several tens of microns, and it limits achieving smaller footprint. By increasing the refractive index contrast which consequently increases modal confinement, more compact and denser PICs are possible. The modal confinement is expressed by the fraction of the mode energy density confined in the waveguide core [8]. In this Letter, low-loss bent silicon nitride waveguide is proposed and experimentally evaluated for visible light applications using 638nm light source. The proposed bent waveguide structure has an adiabatic profile and a waveguide offset for reducing a lateral misalignment loss between the proposed bent and straight waveguides as shown in Fig.1.

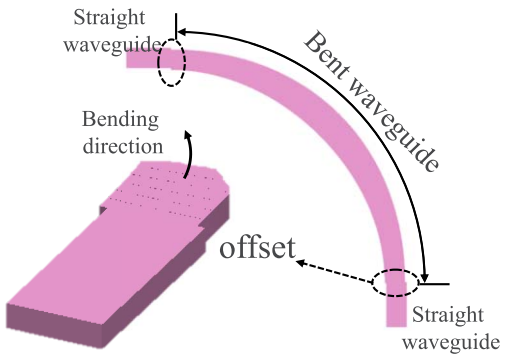


Fig. 1. Proposed bent waveguide employing an adiabatic profile with offsets at the interface between the bent and straight waveguides.

II. Design Approaches

Bends and narrow waveguides have weaker modal confinement which will increase the radiation loss [8]. The radiation loss from a bend can be reduced by increasing the modal confinement. The modal confinement factor represents the fraction of power confined to a volume of the waveguide core. It can be calculated using the film mode matching method with a complex mode solver and perfectly matched layer (PML) boundary conditions. Fig.2 (a) shows the confinement factor as a function of waveguide width when a bending radius is varied. Here, the refractive indices of silicon nitride and oxide are assumed to be ~1.91 and ~1.46 as a core and cladding, respectively. The waveguide thickness is assumed to be 220nm. Fig.2 (b) shows bending losses as a function of waveguide width. It also shows that the wider waveguide width rapidly reduces bending loss in the case of large bending radius. The bending losses were calculated from the imaginary part of the propagation constant [9]. This make us to generate an idea that the characteristics of bending loss can be improved by gradually or/and rapidly varying the waveguide width or the shape of bends. Namely, applying an adiabatic profile on a bent waveguide can reduce the radiation loss. Thus, a sinusoidal

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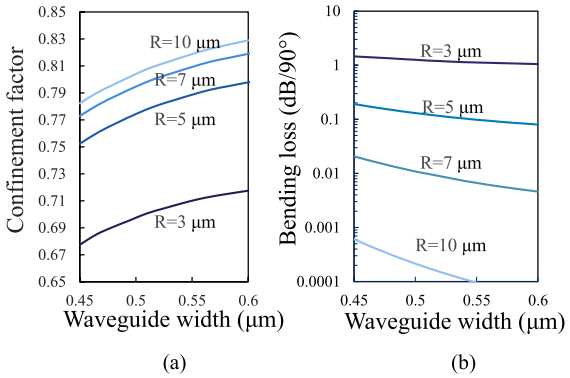


Fig. 2. Confinement factor (a) and bending loss (b) as a function of waveguide width with various bending radii (R).

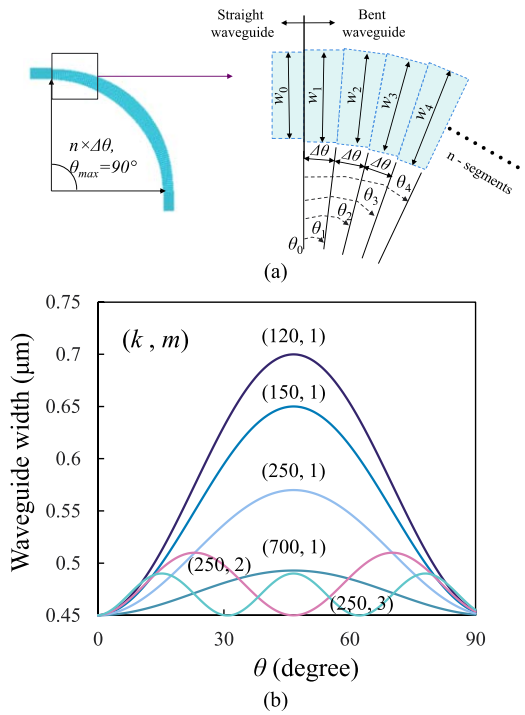


Fig. 3. (a) Design parameter definitions of width (w_n) and angle (θ_n) and (b) Employed adiabatic profiles with k and m parameters. Here, $w_0 = 450$nm.

bend has been used as the adiabatic profile. This profile of an adiabatic bend can be derived by calculating the varied waveguide widths along its length. This varied width of any segment n of the waveguide bend can be expressed as:

$$w_n = w_{n-1} + \frac{1}{k}\sin(mq\theta_n). \quad (1)$$

w and k are the width and the coefficient of width variable, respectively. θ_n is the angle of bend which is corresponding to $n \times \Delta\theta$ [see Fig.3(a)]. $q = 3.819719$. m is an integer which indicates the number of envelops [see Fig.3(b)]. w_0 and θ_0 are the initial straight waveguide width and angle, respectively. Fig.3 (b) shows the plot of (1) and the m and k vary the periodicity and the width of sinusoidal bend, respectively. Bent waveguides have been drawn by (1) and then three dimensional finite difference time domain (3D-FDTD) simulations have been implemented as shown in Fig.4. The simulation setup is that the bend structure drawn by (1) is interconnected with a straight waveguide, which is single mode operational at 638nm, as in- and output waveguides. We designed the channel waveguide with a 450nm-wide and 220nm-thick on the silicon nitride platform. This dimension of the waveguide can support fundamental single transverse electric (TE) and transverse magnetic (TM) modes at longer than 635-nm wavelength. The core and cladding indices in silicon nitride platforms provided by IMEC are 1.902 and 1.46, respectively [6]. Fig.4 (a) is a conventional (non-adiabatic profile) bend when the radius is 5μm. Fig.4 (b) shows that the adiabatic bend has higher transmission power than the conventional bend [Fig.4 (a)]. Among bent structures, the optimal value of m was determined to be one. k value is needed to be optimized for each bending radius. Example structures are shown in Fig.4 and their simulation results with the parameters of k and m are summarized in Table I. Here, we considered TE mode for the design and measurement.

Another method to reduce bending loss is using an offset between straight and bent waveguides [10] as shown in Fig.1. The optical mode has Gaussian-distribution in the core of waveguide and peak power is located at the center of core. The effective index will be increased and simultaneously the power distribution of mode in the core will be shifted from the center to edge of core when a bend is occurred. Consequently, the shape of mode in the bent waveguide is changed as shown in the inset of Fig.5. Scattering occurs when light propagates across the interface between a straight and a bent waveguide. To minimize scattering loss, the offset can be used at the interface between a bent and a straight waveguides. Using 3D-FDTD simulation with varying offsets, we can determine optimal values where the transmission of the fundamental mode across the interface is maximal as shown in Fig.5. A smaller radius needs a larger offset. It means that the change of effective index at the bend will be larger when bend radius is smaller.

We have employed the offset 40nm or more than that on the actual design mask. We considered 5% of the fabrication tolerance, therefore if the offset becomes less than 40nm on the mask the possibility to obtain uncertain effect will be increased. The fabrication tolerance error comes from the difference between the sizes of litho-target structures and other structures when implementing lithography process. When radii are 7μm and 10μm, optimal offsets are near 25nm. However, 40nm-offsets were designed on the mask. The loss difference between using 25nm-offset and 40nm-offset is less than 0.03dB/90°.

Fig.6 shows transmission spectra as a function of k value with different bending radii. The offsets employed here are 80nm, 40nm, 40nm and 40nm when radii are 3μm, 5μm, 7μm, and 10μm, respectively. Optimized k values were found to be 700, 150, 150, and 120 when radii are 3μm, 5μm, 7μm, and 10μm, respectively. Fig.7 (a) shows the electric field plot of adiabatic bend with optimal k and 40nm-offset when the radius is 5μm. The electric field of the conventional bend without offset is shown in Fig.7 (b). The radiation loss can be seen clearly in Fig.7 (b). Calculated bending losses are 0.19dB/90° and 0.39dB/90° at the adiabatic bend with the offset and conventional bend without the offset, respectively. For comparison, the bending loss of adiabatic bend without offset is 0.29dB/90°.

III. Measurement Results and Discussion

The silicon nitride waveguides and bends were fabricated by PECVD (plasma enhanced chemical vapor deposition) method. The platform's propagation loss is

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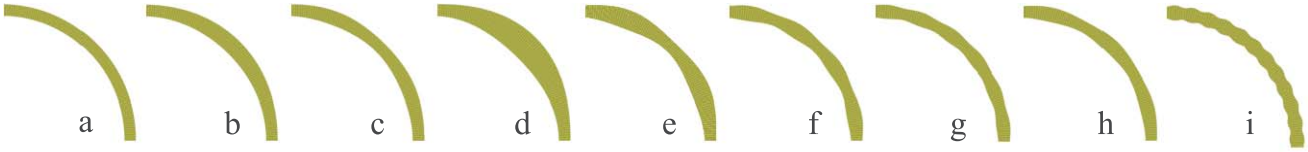


Fig. 4. Examples of adiabatic bent waveguides without offsets when R=5μm. Profile parameters and transmissions are summarized in Table I. The conventional bend is (a) and optimal bend with the adiabatic profile is (b).

TABLE I
EXAMPLES OF VARIOUS BENT SHAPES IN FIG.4 AND THEIR SIMULATION PARAMETERS AND RESULTS

Samples / R=5μm (min. width= w_0)	a	b	c	d	e	f	g	h	i
(k , m)	(inf., 0)	(150, 1)	(150, 2)	(50, 1)	(50, 2)	(50, 4)	(50, 6)	(50, 3)	(20, 12)
Transmission for 638nm	*0.912*	*0.935*	0.916	0.902	0.693	0.881	0.868	0.844	0.845
Widest width	450nm	650nm	550nm	1050nm	750nm	600nm	550nm	650nm	573nm

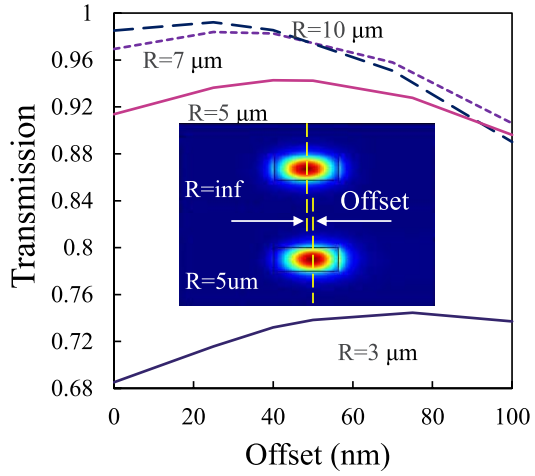


Fig. 5. Optimized offsets with different bending radii. Inset shows lateral shift by bending.

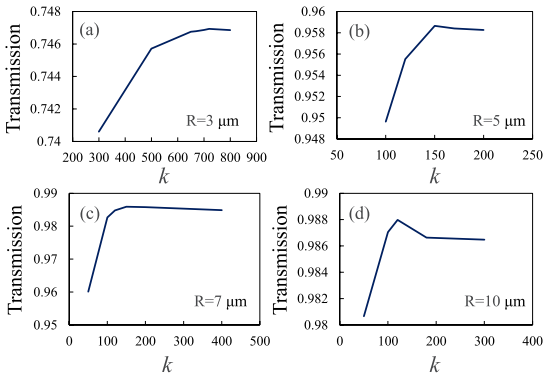


Fig. 6. Optimization of k values when R=3μm (a), 5μm (b), 7μm (c), and 10μm (d). The data are from when using offsets.

~1.0dB/cm with a 638nm light source. The waveguide has 220nm-thick silicon nitride layer with the top and under cladding of 2.0μm and 2.3μm, respectively. Four different bends were tested. 1) Conventional bend without offsets, 2) Conventional bend with offsets, 3) Adiabatic bend without offsets, and 4) Adiabatic bend with offsets. Each waveguide

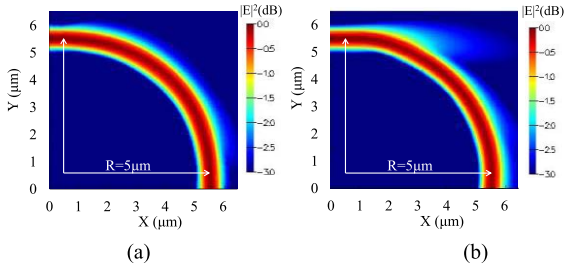


Fig. 7. Electric field plots of bends (a) with adiabatic profile and offset (b) without adiabatic profile and offsets (conventional bend w/o offsets).

layout contains a straight line with several bends. The number of bends is dependent on the radius of bend which varies from 3μm to 10μm. All waveguides have their in- and output-grating couplers. A reference straight waveguide is also included in each test site.

A typical grating coupler measurement setup [11] was used for the banding loss test. The output of a 638nm laser source was connected through a polarization-controller to an input-fiber for using TE mode. The output-fiber was connected to a broadband power detector. The input and output fibers were mounted on a precision tilting stage on three axis automatic-translation stages. The angle of input and output fibers was 10°. Here, in-and output-fibers are high index core fibers, where diameter is ~6μm.

Fig.8 shows measurement results compared with simulation data. As expected, the adiabatic bends with offsets have much smaller bending loss than conventional bends. The results are summarized in Table II.

The bending performance for TM modes will also be improved with the adiabatic bends. 3D-FDTD simulations show that the transmission of TM mode was 80% when using the adiabatic bend (k = 150 and m = 1) with the radius of 5μm, which is optimized for the TE mode of 638nm. The transmission was 77% at the conventional bend with 5μm radius.

An optimal k value may exist for every bending radius as shown in Fig.6. The graded widths of the adiabatic bends can be determined by k values which behave inversely with bending radii. This can be explained by the modal confinement and multimode interference. The modal confinement increment

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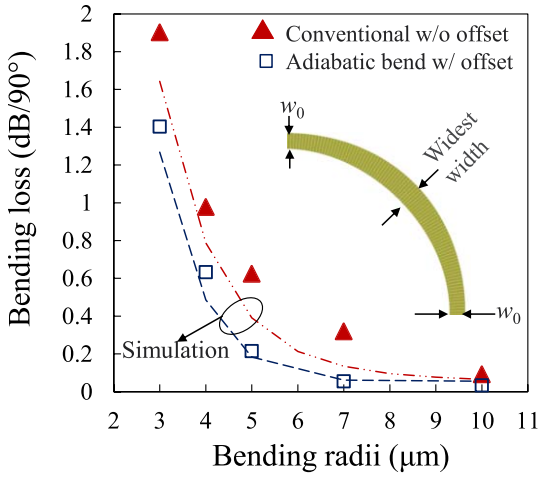


Fig. 8. Measurement results of bending loss vs. Radii. The inset shows an adiabatic bend shape. Optimum parameters are summarized as; k values: 120, 150, 150, 250, 700; *Offsets*: 80, 50, 45, 40, 40nm; *Radii*: 3, 4, 5, 7, 10μm; *Widest widths*: 700, 650, 650, 570, 493nm; respectively, $m = 1$, and $w_0 = 450$nm.

TABLE II

SUMMARY OF MEASURED BENDING LOSSES; AVERAGE VALUES FROM FOUR DIFFERENT DIES

λ=638nm Radius	Adiabatic with offset	Conventional with offset	Adiabatic Only	Conventional Only
10μm	0.034(0.012)	0.040(0.012)	0.124(0.021)	0.091(0.017)
7μm	0.054(0.021)	0.139(0.018)	0.241(0.037)	0.319(0.074)
5μm	*0.215*(0.032)	0.341(0.046)	0.489(0.040)	*0.623*(0.063)
4μm	0.633(0.070)	0.752(0.088)	0.868(0.054)	0.977(0.083)
3μm	1.403(0.133)	1.561(0.211)	1.899(0.235)	1.901(0.163)

*Unit of loss: dB/90°, ( ) standard deviations

TABLE III

SIMULATION PARAMETERS AND RESULTS USING C-BAND SOURCE

λ=1550nm Radius 20μm	Adiabatic offset 80nm, k=50, m=1 (dB/90°)	Conventional offset 80nm (dB/90°)	Adiabatic w/o offset k=50, m=1 (dB/90°)	Conventional w/o offset (dB/90°)
Bending loss	0.529	0.793	0.597	0.852

by radii will be much larger than that by widths varied by k values. In a large bending radius with a small k value, which makes the width wider on an adiabatic bend, the loss from multimode interference will be higher than that from bending even though the wider width on the bend helps reducing bending loss.

All bends used in this design are for 638nm wavelength. However, the design can be employed for telecommunication applications using C-band wavelengths (1530nm-1565nm). Initial simulation using a 1550nm light source shows that the loss characteristics of the adiabatic bent waveguide is also improved. The refractive indices of silicon nitride and oxide were assumed to be 1.862 and 1.45, respectively at 1550nm. The single mode waveguide has the width of 1000nm and the thickness of 400nm. The longer wavelength has weaker confinement and the bending loss is higher than shorter wavelength. The confinement factors of the cases of using 638nm and using 1550nm here are 0.790 and 0.586, respectively. The results of simulations using a 1550nm light source are summarized at the Table III. The optimized offset was 80nm when the radius is 20μm. According to measured and simulated data, we believe that the adiabatic profile can be employed for any bends using universal range of lights.

IV. CONCLUSION

Low-loss bent waveguides composed by an adiabatic profile with an optimized offset have been proposed and demonstrated. The profile gradually increases waveguide width by up to 45° and then decrease up to 90° symmetrically. The modal confinement is increased and bending loss is reduced in this structure simultaneously. As a result of experimental evaluation, the characteristics of bending loss are improved. Bending loss of 0.215dB/90° has been achieved at 5-μm radius. The loss of this bend is ∼0.4 dB/90° lower than a conventional bend with the same radius. We also investigated bends when using a C-band light source theoretically. Less than 0.6dB/90° is showed when employing the adiabatic profile with the bending radius of 20μm. It has also better performance than a conventional bend. Thus, the adiabatic bend can be used for various applications using the wide range of visible and infrared lights. This technology can be a good solution for silicon nitride waveguide platforms with high density and compactness required.

REFERENCES

[1] K. Yamada *et al.*, "High-performance silicon photonics technology for telecommunications applications," *Sci. Technol. Adv. Mater.*, vol. 15, no. 2, p. 024603, 2014.

[2] Y. Li, Y. Zhang, L. Zhang, and A. W. Poon, "Silicon and hybrid silicon photonic devices for intra-datacenter applications: State of the art and perspectives [Invited]," *Photon. Res.*, vol. 3, no. 5, pp. B10–B27, 2015.

[3] B. G. Lee, J. O. Plouchart, A. V. Rylyakov, J. H. Song, F. E. Doany, and C. L. Schow, "Passive photonics in an unmodified CMOS technology with no post-processing required," *IEEE Photon. Technol. Lett.*, vol. 25, no. 4, pp. 393–396, Feb. 15, 2013.

[4] R. Stahl *et al.*, "Microscope-on-chip: Combining lens-free microscopy with integrated photonics," *Proc. SPIE*, Imaging, Manipulation, and Analysis of Biomolecules, Cells, and Tissues XIII, vol. 9328, p. 93281C, Mar. 2015.

[5] B. Jalali and S. Fathpour, "Silicon photonics," *J. Lightw. Technol.*, vol. 24, no. 12, pp. 4600–4615, Dec. 2006.

[6] A. Z. Subramanian *et al.*, "Low-loss singlemode PECVD silicon nitride photonic wire waveguides for 532–900 nm wavelength window fabricated within a CMOS pilot line," *IEEE Photon. J.*, vol. 5, no. 6, Dec. 2013, Art. no. 2202809.

[7] Y. A. Vlasov and S. J. McNab, "Losses in single-mode silicon-on-insulator strip waveguides and bends," *Opt. Exp.*, vol. 12, no. 8, pp. 1622–1631, Apr. 2004.

[8] T. D. Visser, B. Demeulenaere, J. Haes, D. Lenstra, R. Baets, and H. Blok, "Confinement and modal gain in dielectric waveguides," *J. Lightw. Technol.*, vol. 14, no. 5, pp. 885–887, May 1996.

[9] D. Dai and S. He, "Analysis of characteristics of bent rib waveguides," *J. Opt. Soc. Amer. A*, vol. 21, no. 1, pp. 113–121, 2004.

[10] T. Kitoh, N. Takato, M. Yasu, and M. Kawachi, "Bending loss reduction in silica-based waveguides by using lateral offsets," *J. Lightw. Technol.*, vol. 13, no. 4, pp. 555–562, Apr. 1995.

[11] J. H. Song, R. A. Budd, B. G. Lee, C. L. Schow, and F. R. Libsch, "Focusing grating couplers in unmodified 180-nm silicon-on-insulator CMOS," *IEEE Photon. Technol. Lett.*, vol. 26, no. 8, pp. 825–828, Apr. 15, 2014.

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